STUDENT ASSIGNMENT BRIEF

Hackathon

Problem statement assignments

37 teams / 9 assigned problem statements

Team rosters, project scope and expected deliverables

Theme and problem statement order

Theme	Problem statements	Teams
T1 Climate & Disaster Resilience	PS-01 CoolRoute	5
T2 Smart Cities & Public Safety	PS-02 Vision Zero	4
T3 Healthcare & Wellbeing	PS-03 MaaSuraksha; PS-04 VaxReach	8
T4 Education & Employability	PS-05 SignBridge; PS-06 ShikshaSetu	7
T5 Agriculture & Rural Innovation	PS-07 KisanMitra	5
T6 Sustainability & Waste Management	PlasticWatch; ReLoop (not assigned)	0
T7 Cybersecurity & Digital Trust	PS-08 Phishing Domain Detection	4
T8 Civic Tech & Digital Governance	PS-09 CivicFix	4

PS-01 to PS-09 are internal allocation labels. T6 topics appear for reference only.

Project and submission guidance

Assignments	Each team has one assigned PS. All 37 teams received their first-choice theme.
Working prototype	Build a working prototype that covers your PS deliverables. Show the required explanations, confidence values and safeguards.
Demo video	Record a demo video showing the problem, the main user workflow and the prototype running with sample inputs and outputs.
GitHub repository	Provide the source code in an accessible GitHub repository. Include a README with setup steps, dependencies, data sources and the demo video link.
Data and setup	Use the offline starter pack. Prepare synthetic data where stated and clearly label it. Follow the instructions in your PS folder.

PPT submission timeline

1. Initial solution PPT

Due by 11:59 PM (IST) on the day of PS assignment

Submit a PPT with your team name and PS ID. Explain the problem, your proposed solution, the main features, technology and data you plan to use, and how you will build the prototype.

2. Final PPT for the finale

Present your completed prototype. Include demo screenshots, a working demo video link, the GitHub repository link, results and key limitations. Make sure reviewers can open the links.

Keep the same PPT structure where possible. You may update the approach and content as you improve the project for the finale.

PS-01 CoolRoute

Heat-aware routing

Problem description

Recommend walking or work routes that reduce heat exposure, while explaining the trade-off against travel time. Heat exposure varies with time of day, surface temperature, shade, humidity and access to water or rest points.

Scope

One campus or city zone. Compare at least two routes.

Assigned teams (5)

Team	Team leader
Cache me if you can	Suchismita Singh
EliteX	Subham Prasad
HACK HORIZON	Ujjayini Sahoo
Gypsy Danger	Devashrit Sarangi
Pacific Monkeys	Sai Soham Pattanaik

CoolRoute: deliverables

Expected deliverables

1. A working prototype for one campus or city zone.
2. Comparison of at least two routes using an open, explainable heat score.
3. A map or display of shade and water points.
4. Rerouting when a simulated environmental condition changes.

Data to use

Offline NASA weather sample. Prepare synthetic route segments, shade and water-point data.

Review focus

Show heat exposure alongside travel time. Explain the score and demonstrate how a changed condition alters the route.

Key constraint

Keep the heat score transparent and the demonstration within the chosen zone.

PS-02 Vision Zero

Road risk and near-miss intelligence

Problem description

Identify dangerous road locations from near-misses and traffic-conflict patterns before crashes occur. Analyse recorded or simulated traffic and turn risk indicators into useful information for road engineers.

Scope

Recorded or simulated traffic. At least one near-miss indicator.

Assigned teams (4)

Team	Team leader
Team shannon	Bibhesh Nayak
Vigilante Coders	Ankita Dutta
AstraX	Aditya Gupta
CUTE CODERS	AAKASH KUMAR

Vision Zero: deliverables

Expected deliverables

1. Recorded or simulated traffic-data input.
2. Vehicle and pedestrian detection.
3. Calculation and explanation of at least one near-miss indicator.
4. A map of road-risk hotspots.
5. A report or summary for road engineers.

Data to use

Prepare synthetic traffic scenes and tracks with vehicle/pedestrian labels, timestamps and positions.

Review focus

Demonstrate detection, explain the near-miss indicator and show how mapped hotspots support engineering review.

Key constraint

No face identification and no assignment of fault.

PS-03 MaaSuraksha

Offline-first maternal risk screening

Problem description

Give village health workers an offline tool to record maternal danger signs, explain risk and support referral. Capture observations such as blood pressure, haemoglobin, bleeding, swelling or reduced fetal movement.

Scope

Android app or progressive web app (PWA), with offline capture and later sync.

Assigned teams (4)

Team	Team leader
Knight Mares	Baren Tripathy
SoulMatrix	Pratyush Kumar Satapathy
BYTIX	Om prakash Nahak
RAYORA	Satyajit patra

MaaSuraksha: deliverables

Expected deliverables

1. An Android app or PWA with offline case capture.
2. Multilingual or voice input.
3. Risk levels with understandable reasons from entered danger signs.
4. Later synchronisation to a supervisor dashboard.
5. A clear workflow supporting referral by trained health workers.

Data to use

Offline UCI Maternal Health Risk CSV. Prepare synthetic cases for danger signs missing from the dataset.

Review focus

Demonstrate offline entry and later sync. Show understandable risk reasons and a usable referral workflow.

Key constraint

Support trained health workers. The tool must not claim to diagnose.

PS-04 VaxReach

Immunisation gap detection and outreach planning

Problem description

Find children and localities likely to miss vaccinations and plan targeted outreach. Overall coverage may hide missed children because of migration, travel distance or repeatedly missed sessions.

Scope

De-identified or synthetic immunisation-session data. Logistics support only.

Assigned teams (4)

Team	Team leader
Rounak	Rounak padhi
NovaForge	Priyabrata Das
ZeroTrace	Shatabdi Panigrahy
TeamGO	Yudhajit Jena

VaxReach: deliverables

Expected deliverables

1. Import of de-identified or synthetic session records.
2. Flags for missed doses and doses that are due.
3. A map of locality-level coverage gaps.
4. Prioritised outreach lists and a proposed session or route plan.
5. Fairness and coverage metrics, with an audit trail for each priority decision.

Data to use

Prepare synthetic vaccination records: child ID, locality, dose, due date, given date and session.

Review focus

Connect missed/due-dose flags to an actionable outreach plan. Explain priorities and measure fairness and coverage.

Key constraint

Keep the system focused on logistics. It must not make clinical decisions.

PS-05 SignBridge

Indian Sign Language communication bridge

Problem description

Translate a small, defined set of Indian Sign Language (ISL) gestures into text or speech, or in the reverse direction. Focus on reliable communication for one setting, such as a classroom, help desk or hospital counter.

Scope

Starter scope: A, B and C recognition for a spelling-help use case.

Assigned teams (4)

Team	Team leader
Team Solaris	Subham Rout
The Innovatrix	Aryan Mohanty
Nexbyte	Sankalpa Pattnayak
GODSE	K PURUSOTAM ACHARY

SignBridge: deliverables

Expected deliverables

1. A prototype for one clearly defined use case.
2. A fixed, limited vocabulary of ISL gestures.
3. Gesture-to-text/speech or text/speech-to-gesture functionality.
4. Confidence display for predictions.
5. Accuracy evaluation on held-out samples.

Data to use

Offline ISL alphabet images: A, B and C. Use a spelling-help demo; this subset does not cover word signs.

Review focus

State the vocabulary and test the selected communication workflow. Report held-out accuracy and show confidence.

Key constraint

Keep claims within the tested vocabulary. Do not claim universal ISL translation.

PS-06 ShikshaSetu

Explainable early-intervention platform

Problem description

Identify students who may be at academic or engagement risk early, explain why and suggest interventions for mentors. Use attendance and assessment information to support timely action and track changes over time.

Scope

Explainable risk bands with mentor feedback and at least three intervention options.

Assigned teams (3)

Team	Team leader
Hack verse	Ayush bardhan panigrahy
Code Craft	Joydip Mahata
Algominds	Sasmita Kar

ShikshaSetu: deliverables

Expected deliverables

1. A dashboard showing student risk bands.
2. Feature-level explanations for each risk estimate.
3. At least three intervention options.
4. A mentor feedback mechanism and tracking of risk over time.
5. Fairness checks and safeguards against permanent student labels.

Data to use

Offline UCI Student Performance CSVs. Prepare synthetic mentor feedback and follow-up records.

Review focus

Show explanations, practical interventions and mentor feedback. Demonstrate changes in risk over time and fairness checks.

Key constraint

Do not present correlation as causation or permanently label students.

PS-07 KisanMitra

Crop risk decision support for small farmers

Problem description

Combine crop imagery or symptoms, growth stage, weather and market information into a few clear next actions. Help farmers understand which risks matter now, within one crop and one region.

Scope

One crop and one region, with clear uncertainty in the advice.

Assigned teams (5)

Team	Team leader
DietCode	Akash Pattnaik
QUANTUM BURNER 2.0	Chandan Sahoo
Power Rangers	Reshma Pradhan
SYNQ	Saijeetan Moharana
Code HEX	Shakti Prasad Nayak

KisanMitra: deliverables

Expected deliverables

1. Image or symptom-based input, with crop-stage context.
2. Crop-risk classification.
3. Weather-aware advice.
4. A simple market or pricing view.
5. Clear, actionable recommendations that communicate uncertainty.

Data to use

Offline tomato-leaf images. Prepare synthetic crop-stage, weather and mandi-price records.

Review focus

Show how crop risk, weather and market context affect the recommended actions for the selected crop and region.

Key constraint

State uncertainty and avoid guaranteed pesticide claims.

PlasticWatch

Plastic waste hotspot detection

Problem description

Detect likely plastic accumulation from geotagged images, merge duplicate reports and rank cleanup locations. Help prioritise waste near drains, riverbanks, markets and roadsides.

Scope

Reference topic from the guide. No teams assigned in the final workbook.

Not assigned

No teams are allocated to this topic in the final assignment workbook.

PlasticWatch: deliverables

Expected deliverables

1. Plastic-waste classification from sample images and geotagged reports.
2. A map of detections.
3. Duplicate-report merging.
4. Hotspot ranking by severity, recurrence and proximity to drains or water bodies.
5. Confidence display.

Data to use

Offline TrashNet image subset replaces TACO. Prepare synthetic geotags, repeats and hotspot severity.

Review focus

Demonstrate classification, mapping and duplicate merging. Explain hotspot rankings and display confidence.

Key constraint

Citizen photos must not serve as proof of who is responsible.

ReLoop

Material recovery pathways and e-waste handover

Problem description

Identify a discarded item, recommend reuse, repair or recycling and match it with a verified handler. Help households find the right channel for reusable items and e-waste, and track handover.

Scope

Reference topic from the guide. No teams assigned in the final workbook.

Not assigned

No teams are allocated to this topic in the final assignment workbook.

ReLoop: deliverables

Expected deliverables

1. Item identification from an image or text.
2. A recommendation for reuse, repair or recycling.
3. Nearby approved or verified handler options.
4. Handover-status recording.
5. A demonstration that includes an e-waste case.

Data to use

Offline TrashNet sample for common items. Prepare synthetic e-waste, handler and handover records.

Review focus

Demonstrate item identification, a suitable recovery pathway, verified handler selection and recorded handover status.

Key constraint

Never direct hazardous items to unverified handlers.

PS-08 Phishing Domain Detection

AI/ML detection of domains imitating genuine websites

Problem description

Build an automated AI tool that detects look-alike phishing domains among newly registered websites. Phishing sites imitate genuine login pages to steal credentials or deliver malware.

Scope

Offline baseline with simulated new-domain cases. Live WHOIS access is not required.

Assigned teams (4)

Team	Team leader
Hacksmiths	Aurovind Maharana
VectorX	Pratyush Priyadarsi
Ctrl C + Ctrl V	Subham Mekap
NextGen Coder	Pabitra Mohan Sahu

Phishing Domain Detection: deliverables

Expected deliverables

1. Automated detection of look-alike phishing domains.
2. Offline labelled URL data and synthetic look-alike domain pairs.
3. Similarity analysis against genuine domains, using content/code or page images as feasible.
4. Probability or closeness scores.
5. Fast detection with usable output in flexible formats.

Data to use

Offline PhiUSIIL URL sample replaces WHOIS. Prepare synthetic look-alike domain pairs for similarity tests.

Judging dimensions

Source-stated judging dimensions: closeness probability scores, detection speed, ease of use and flexible output formats.

Key constraint

Explain the similarity evidence and make probability or closeness scores visible.

PS-09 CivicFix

Grievance triage, deduplication and accountability

Problem description

Classify municipal complaints, group duplicate reports, score priority transparently and track service deadlines. Support complaints about issues such as potholes, waste, streetlights and drains, even when location details are vague.

Scope

At least five issue types, with text or image input and an administrator dashboard.

Assigned teams (4)

Team	Team leader
Aryan Panigrahy	Aryan Subhajit Panigrahy
ECLIPSE	SPANDAN SINHA
The Phantom Stack	Susobhana Mohanty
git push pray	Adarsh Kumar Singh

CivicFix: deliverables

Expected deliverables

1. Complaint input accepting text or images.
2. Support for at least five issue types.
3. Location mapping for reported issues.
4. Duplicate detection and grouping.
5. Transparent priority scoring and an administrator dashboard with SLA deadlines.

Data to use

Prepare synthetic complaints across five issue types, with locations, duplicates, status and SLA deadlines.

Review focus

Demonstrate complaint classification, location mapping and duplicate grouping. Show priority reasons and service-level agreement (SLA) tracking.

Key constraint

Do not silently reject complaints or hide how priority is assigned.